

Aggregate Demand and Aggregate Supply

Mbak 9515: Economics

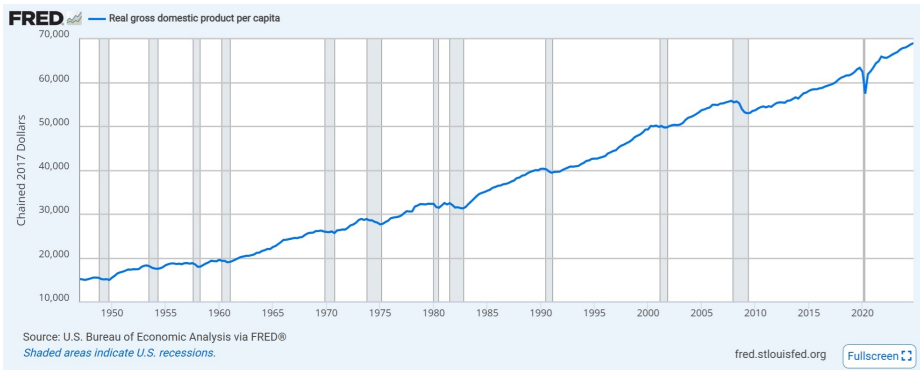
Sungjun Huh

Konkuk University

Readings

- Chapter 15. Aggregate Demand and Aggregate Supply
 - ▶ Economic fluctuations
 - ▶ The Aggregate Demand Curve
 - ▶ The Aggregate Supply Curve
 - ▶ Causes of Economic Fluctuations

In the Long-run



In the Short-run

Recessions

- Some economists compare two large recessions: the Great Depression and the Great Recession
- Recessions are costly and very painful to the people affected

	Output	Consumption	Investment	Employment
Great Recession	-7.2	-5.4	-33.5	-6.7
Average post WW2	-4.4	-2.1	-17.8	-3.8

Source: Christiano, Motto, and Rostagno (2003)

In the Short-run

Recessions

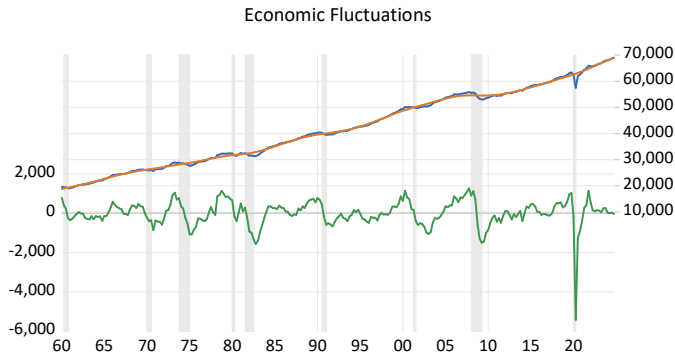
- Some economists compare two large recessions: the Great Depression and the Great Recession
- Recessions are costly and very painful to the people affected.

	Output	Consumption	Investment	Employment
Great Recession	-7.2	-5.4	-33.5	-6.7
Average post WW2	-4.4	-2.1	-17.8	-3.8
Great Depression	-36	-23	-69	-27

Source: Christiano, Motto, and Rostagno (2003)

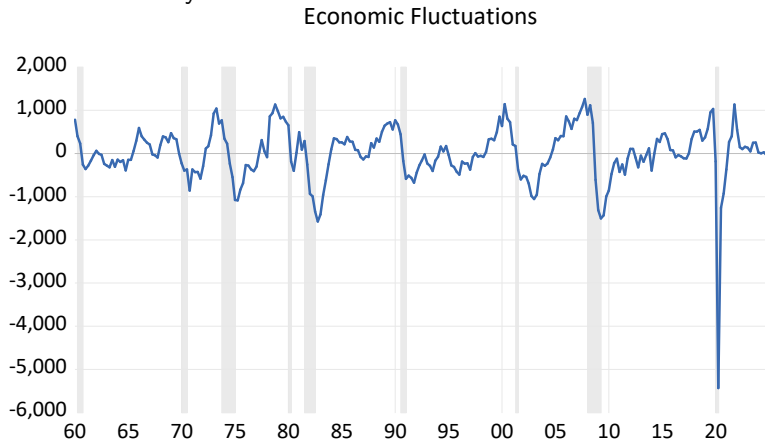
Economic Fluctuations

- An economy does not always grow steadily
 - ▶ Recessions (contractions): slowdowns in economic growth
 - ★ depression: a particularly severe or protracted recession
 - ▶ Expansions: rapid economic growth
 - ★ boom: a particularly strong and protracted expansion



Economic Fluctuations

- Peak: the beginning of a recession, the high point of economic activity prior to a downturn
- Trough: the end of a recession, the low point of economic activity prior to a recovery



Economic Fluctuations

- Economic Fluctuations are *Irregular and Unpredictable*
 - ▶ business cycles or cyclical fluctuations (**No!**)
 - ▶ fluctuations in economic activity do not follow a regular and predictable pattern
- Most Macroeconomic Quantities Fluctuate *Together*
 - ▶ corporate profits, investment, consumption, retail sales, home sales, etc.
 - ▶ “procyclical” variables figure
- As Output *Falls*, Unemployment *Rises*
 - ▶ the unemployment rate rises considerably during recessions
 - ▶ “countercyclical” variables figure
 - ▶ the unemployment rate never falls to zero; instead, it fluctuates around 5 or 6 percent figure

Short-Run Economic Fluctuations

- Classical dichotomy
 - ▶ Separation of variables into: Real variables and Nominal variables
- Monetary neutrality
 - ▶ Changes in the money supply affect nominal variables
 - ▶ Do not affect real variables
- Classical theory holds *in the long-run*
 - ▶ Changes in money supply
 - ▶ Affect prices, and other nominal variables
 - ▶ Do not affect real GDP, unemployment, or other real variables

Short-Run Economic Fluctuations

- *in the short-run*

- ▶ Assumption of monetary neutrality: no longer appropriate
- ▶ Real and nominal variables are highly intertwined
- ▶ Changes in the money supply
 - ★ **Can** temporarily push real GDP away from its long-run trend

Short-Run Economic Fluctuations

- **AD-AS model**

- ▶ Model of aggregate demand (AD) and aggregate supply (AS)
- ▶ Most economists use it to explain short-run fluctuations in economic activity
 - ★ Around its long-run trend

Short-Run Economic Fluctuations

- **Aggregate-demand curve**

- ▶ Shows the quantity of goods and services that households, firms, the government, and customers abroad
- ▶ Want to buy *at each price level*
- ▶ Downward sloping

Basic Economy

Firms don't expect consumers to spend as much as in the last year

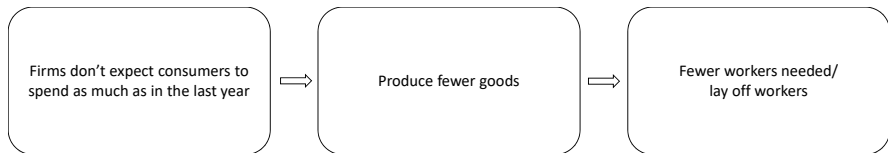
Basic Economy

Firms don't expect consumers to spend as much as in the last year

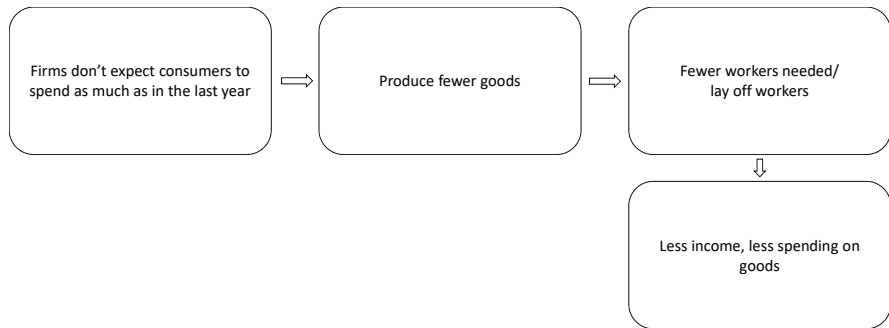


Produce fewer goods

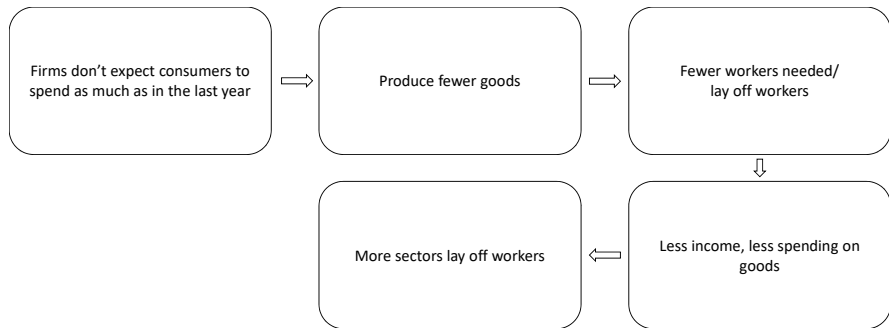
Basic Economy



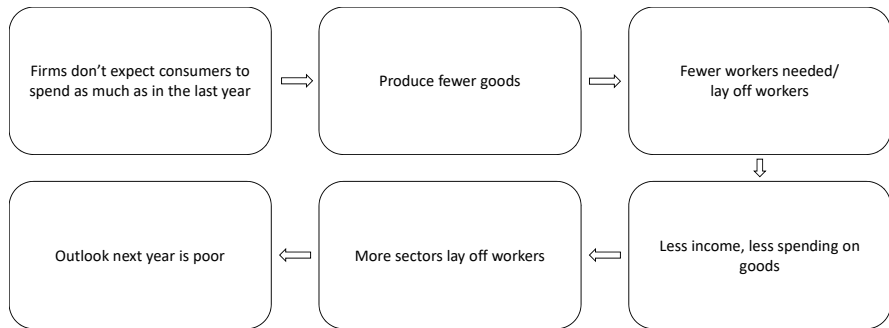
Basic Economy



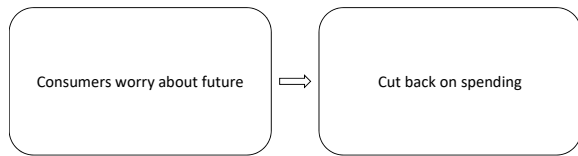
Basic Economy



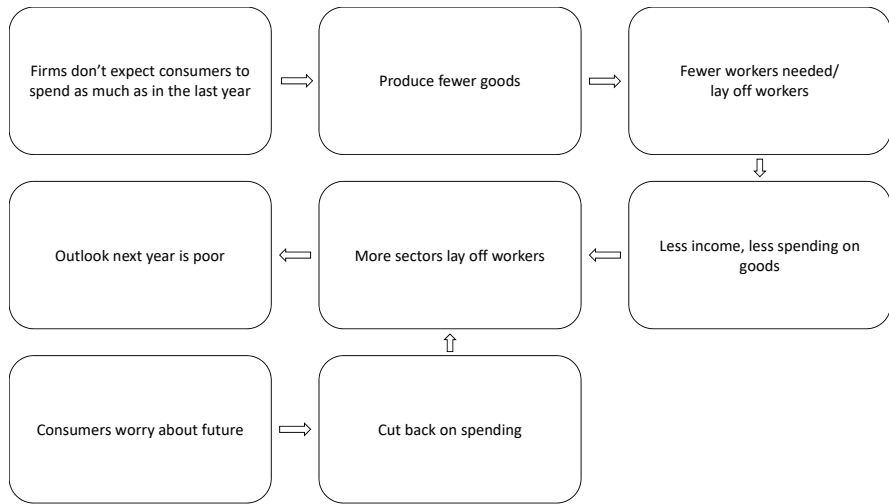
Basic Economy



Basic Economy



Basic Economy



Short-Run Economic Fluctuations

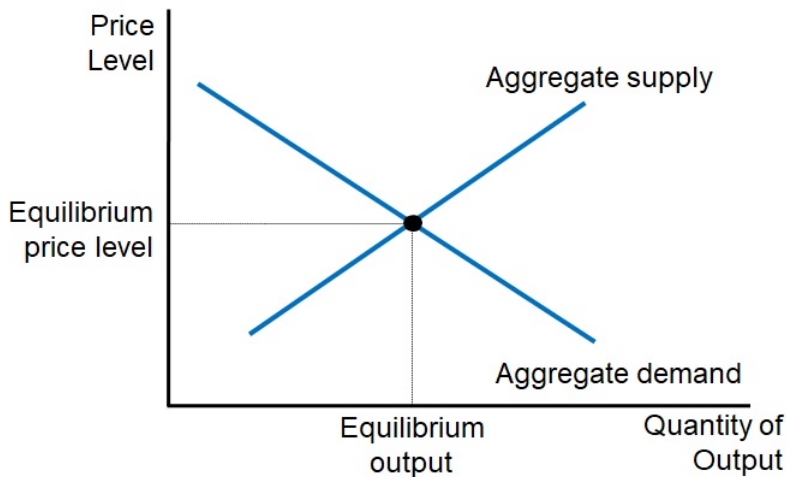
- In the short run, aggregate output is determined by *insufficient spending (demand)*
- Three terms that mean the same thing:
 - ▶ Aggregate expenditure
 - ▶ Total spending
 - ▶ Aggregate demand
- All three terms refer to the total amount that people in the economy plan to buy (or spend).

Short-Run Economic Fluctuations

- **Aggregate-supply curve**

- ▶ Shows the quantity of goods and services that **firms** choose to produce and sell
- ▶ At each price level
- ▶ Upward sloping

Aggregate Demand and Aggregate Supply



The Aggregate-Demand Curve

- Total spending on final goods and services

$$Y = C + I + G + NX$$

- Consumption (C): spending on households on final goods and services
 - ▶ food, clothes, automobiles, furniture
- Investment (I): spending by domestic firms on new capital goods
 - ▶ office buildings, factories, equipment, new houses
- Government purchases (G): purchases by federal, state, and local governments
 - ▶ new schools, hospitals, military hardware
- Net Exports (NX): exports minus imports
 - ▶ Exports: sales of domestically produced goods and services to foreigners
 - ▶ Imports: purchases by domestic residents of goods and services produced abroad

The Aggregate-Demand Curve

- Price level and consumption (C): the wealth effect
- Decrease in price level
 - ▶ Increase in the real value of money
 - ▶ Consumers are wealthier
 - ▶ Increase in consumer spending
 - ▶ Increase in quantity demanded of goods and services

The Aggregate-Demand Curve

- Price level and investment (I): the interest-rate effect
- Decrease in price level
 - ▶ Increase in the real value of money
 - ▶ Consumers are wealthier
 - ▶ Decrease money demand
 - ▶ Therefore interest rate falls as the supply curve in loanable funds market shifts outward
 - ▶ Increase spending on investment goods
 - ▶ Increase in quantity demanded of goods and services

The Aggregate-Demand Curve

- Price level and net exports (NX): the exchange-rate effect
- Decrease in U.S. price level
 - ▶ Decrease in the real exchange rate
 - ▶ U.S. dollar depreciates
 - ▶ Stimulates U.S. net exports
 - ▶ Increase in quantity demanded of goods and services

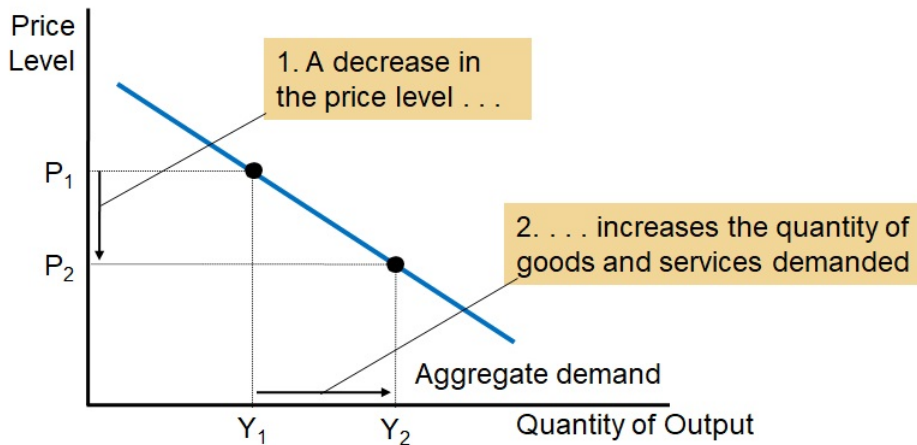
The Aggregate-Demand Curve

- Summary: a fall in price level
 - ▶ Increases quantity of goods and services demanded
 - ▶ Because:
 - ★ Consumers are wealthier: stimulates the demand for consumption goods
 - ★ Interest rates fall: stimulates the demand for investment goods
 - ★ Currency depreciates: stimulates the demand for net exports
- ▶ *We take G as given*

The Aggregate-Demand Curve

- Summary: a rise in price level
 - ▶ Decreases the quantity of goods and services demanded
 - ▶ Because:
 - ★ Consumers are poorer: depress consumer spending
 - ★ Higher interest rates fall: depress investment spending
 - ★ Currency appreciates: depress net exports

The Aggregate-Demand Curve



The Aggregate-Demand Curve

- The AD curve might shift:
 - ▶ Changes in consumption, C
 - ▶ Changes in investment, I
 - ▶ Changes in government purchases, G
 - ▶ Changes in net exports, NX

The Aggregate-Demand Curve

- Changes in consumption, C
 - ▶ Events that change how much people want to consume *at a given price level*
 - ▶ e.g.) Changes in taxes, wealth
 - ▶ Increase in consumer spending
 - ▶ Aggregate-demand curve: shift right

The Aggregate-Demand Curve

- Changes in investment, I
 - ▶ Events that change how much firms want to invest *at a given price level*
 - ▶ e.g.) Better technology, Tax policy, Money supply
 - ▶ Increase in investment
 - ▶ Aggregate-demand curve: shift right

The Aggregate-Demand Curve

- Changes in government purchases, G
 - ▶ Policy makers – change government spending *at a given price level*
 - ▶ e.g.) Build new roads
 - ▶ Increase in government purchases
 - ▶ Aggregate-demand curve: shift right

The Aggregate-Demand Curve

- Changes in net exports, NX
 - ▶ Events that change net exports for *a given price level*
 - ▶ e.g.) Recession in Europe, International speculators – change in exchange rate
 - ▶ Increase in net exports
 - ▶ Aggregate-demand curve: shift right

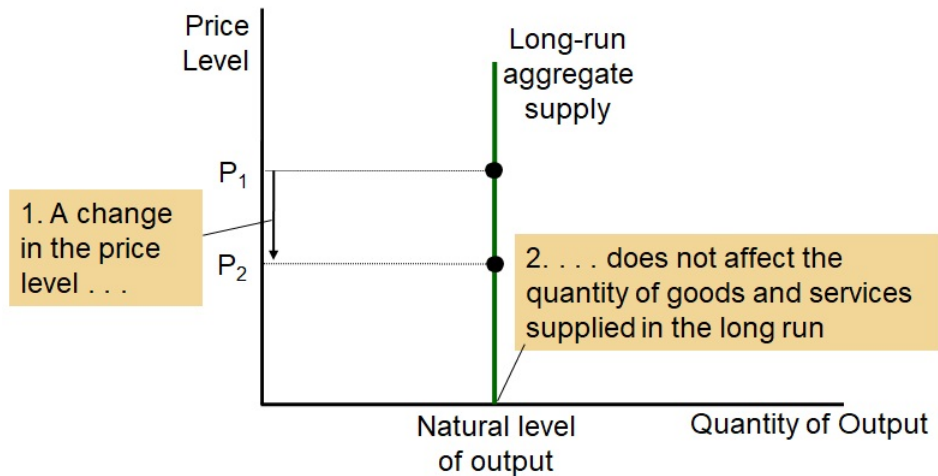
The Aggregate-Demand Curve: Summary

- Why Does the Aggregate-Demand Curve Slope **Downward**?
 - ▶ *The Wealth Effect*: A lower price level increases real wealth, which stimulates spending on consumption.
 - ▶ *The Interest-Rate Effect*: A lower price level reduces the interest rate, which stimulates spending on investment.
 - ▶ *The Exchange-Rate Effect*: A lower price level causes the real exchange rate to depreciate, which stimulates spending on net exports.

The Aggregate-Supply Curve

- Long run aggregate-supply curve is **vertical**, LRAS
 - ▶ Price level does not affect the long-run determinants of GDP:
 - ▶ Supplies of labor, capital, and natural resources
 - ▶ Available technology
- Short run
 - ▶ Aggregate-supply curve is **upward** sloping

The Aggregate-Supply Curve



The Aggregate-Supply Curve

- Natural level of output
 - ▶ Production of goods and services that an economy achieves *in the long run*
- When unemployment is at its **normal rate**
- Potential output
- Full-employment output

The Aggregate-Supply Curve

- The LRAS curve might shift:
 - ▶ Any change in natural level of output
 - ▶ Changes in labor
 - ▶ Changes in capital
 - ▶ Changes in natural resources
 - ▶ Changes in technological knowledge

The Aggregate-Supply Curve

- Changes in labor
 - ▶ Quantity of labor – increases
 - ★ Aggregate-supply curve: shifts right
 - ▶ Natural rate of unemployment – increases
 - ★ Aggregate-supply curve: shifts left
- Changes in capital
 - ▶ Capital stock – increase
 - ▶ Aggregate-supply curve: shifts right
 - ▶ Physical and human capital

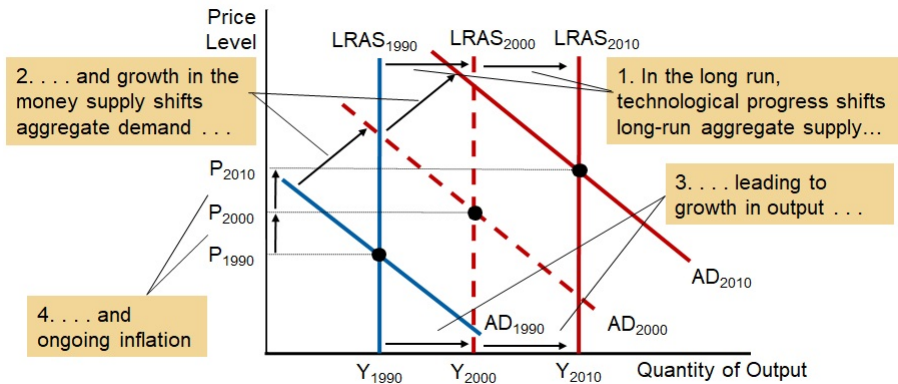
The Aggregate-Supply Curve

- Changes in natural resources
 - ▶ New discovery of natural resource
 - ▶ Aggregate-supply curve: shifts right
 - ▶ Weather
 - ▶ Availability of natural resources
- Changes in technology
 - ▶ New technology, for given labor, capital and natural resources
 - ▶ Aggregate-supply curve: shifts right

Long-Run Growth and Inflation

- Long run: both AD and LRAS curves shift
- Continual shifts of LRAS curve to right
 - ▶ Technological progress
- AD curve shifts to right
 - ▶ Monetary policy
 - ▶ The Fed increases money supply over time
- Result:
 - ▶ Continuing growth in output
 - ▶ Continuing inflation

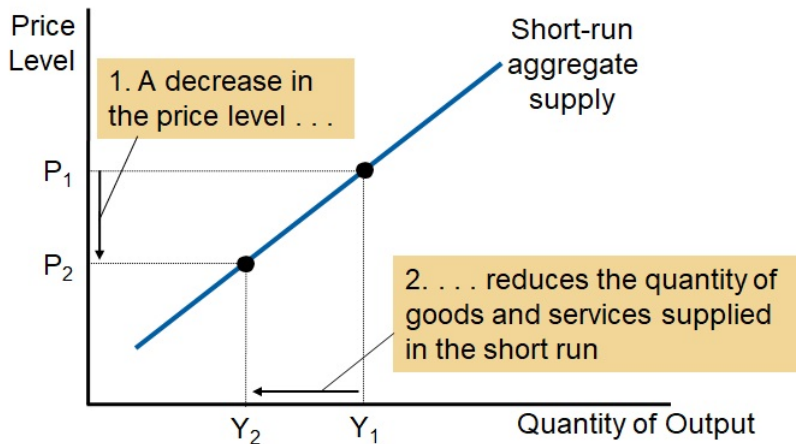
Long-Run Growth and Inflation



The Aggregate-Supply Curve

- Aggregate supply curve slopes *upward* in the short-run:
- Price level affects the economy's output
 - ▶ Increase in overall level of prices in economy
 - ▶ Tends to raise the quantity of goods and services supplied
 - ▶ Decrease in level of prices
 - ▶ Tends to reduce quantity of goods and services supplied

The Aggregate-Supply Curve



The Aggregate-Supply Curve

- Theories that explain why the AS curve slopes upward in short-run:
 - ▶ Sticky-wage theory
 - ▶ Sticky-price theory
 - ▶ Misperceptions theory

The Aggregate-Supply Curve

- Sticky-wage theory
 - ▶ Nominal wages - slow to adjust to changing economic conditions
 - ▶ Long-term contracts: workers and firms
 - ▶ Slowly changing social norms
 - ▶ Notions of fairness - influence wage setting
- ▶ Nominal wages - based on **expected** prices
- ▶ *Don't respond immediately* when actual price level – different from what was expected

The Aggregate-Supply Curve

- Sticky-wage theory
- If price level $<$ expected
 - ▶ Firms – incentive to produce less output
- If price level $>$ expected
 - ▶ Firms – incentive to produce more output

The Aggregate-Supply Curve

- Sticky-price theory
- Prices of some goods and services
 - ▶ Slow to adjust to changing economic conditions
 - ▶ Menu costs: costs to adjusting prices
 - ▶ Dunkin Donuts sets its price at \$1.49
 - ▶ the price level, reflecting mainly the prices of other goods and services, turns out to be unexpectedly high
 - ▶ A cup of coffee now looks “cheap” to consumers
 - ▶ More will visit Dunkin Donuts and buy coffee

The Aggregate-Supply Curve

- Misperceptions theory

- ▶ Changes in the overall price level
 - ▶ Can temporarily mislead suppliers
-
- ▶ Suppose, at the beginning of the year, everyone expects the price level P to equal 100.
 - ▶ But instead, the price level rises to $P = 110$
 - ▶ This means that, on average, the prices of all goods and services have risen by 10 percent.
 - ▶ In the short-run, however, individual firms may mistakenly believe that it is really just the price of their particular good that is rising
 - ▶ Believing that it is an especially good time to produce, those firms will hire more workers
-
- ▶ Changes in relative prices
 - ▶ Suppliers - respond to changes in level of prices
 - ▶ Change - quantity supplied of goods and services

The Aggregate-Supply Curve

- Quantity of output supplied = Natural level of output + α (Actual price level – Expected price level)
 - ▶ Where α - number that determines how much output responds to unexpected changes in the price level

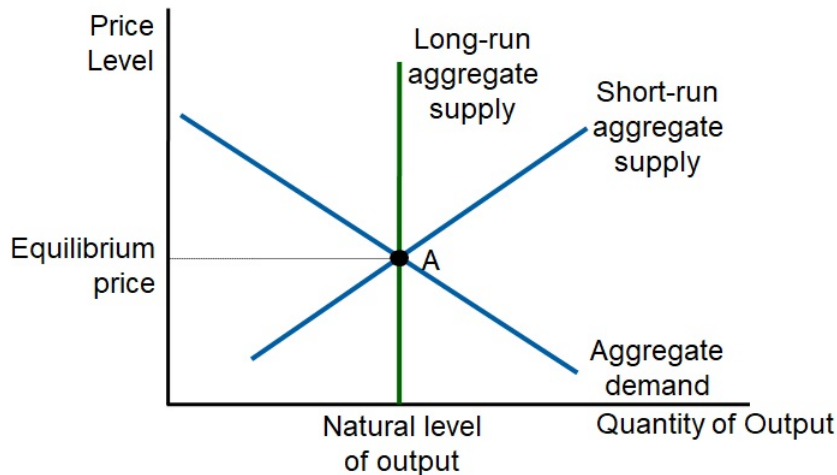
The Aggregate-Supply Curve: Summary

- Why Does the Aggregate-Supply Curve Slope **Upward**?
 - ▶ The Sticky-Wage Theory: An unexpectedly low price level raises the real wage, which causes firms to hire fewer workers and produce a smaller quantity of goods and services.
 - ▶ The Sticky-Price Theory: An unexpectedly low price level leaves some firms with higher than desired prices, which depresses their sales and leads them to cut back production.
 - ▶ The Misperceptions Theory: An unexpectedly low price level leads some suppliers to think their relative prices have fallen, which induces a fall in production.

Causes of Economic Fluctuations

- Assumption
 - ▶ Economy begins in long-run equilibrium
- Long-run equilibrium:
 - ▶ Intersection of AD and LRAS curves
 - ▶ Natural level of output
 - ▶ Actual price level
- Intersection of AD and short-run AS curve
 - ▶ Expected price level = Actual price level

Causes of Economic Fluctuations



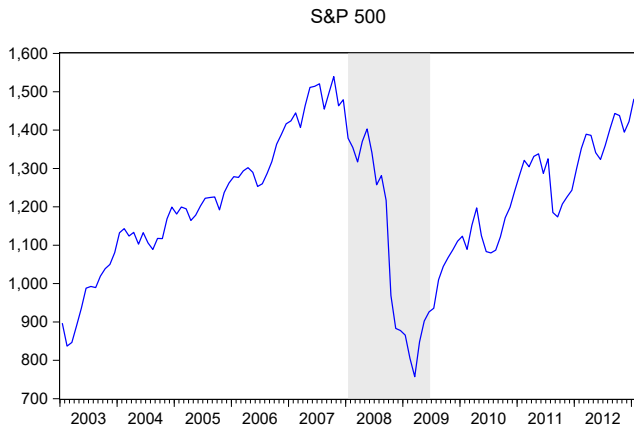
Causes of Economic Fluctuations

- Shift in aggregate demand
 - ▶ Wave of pessimism: AD shifts left
 - ▶ Short-run
 - ★ Output falls, Price level falls
 - ▶ Long-run
 - ▶ Intersection of AD and LRAS curves
 - ▶ Short-run aggregate-supply curve shifts right
 - ▶ Output – natural level, Price level – falls

Example: A Fall in Autonomous Consumption

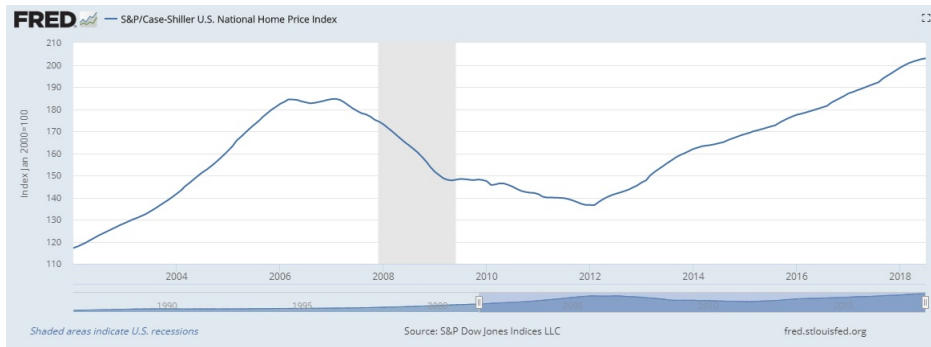
- A fall in consumption not caused by a fall in output.
- A decline in the intercept of the consumption function ($\bar{C}$).

Stock Market Crash

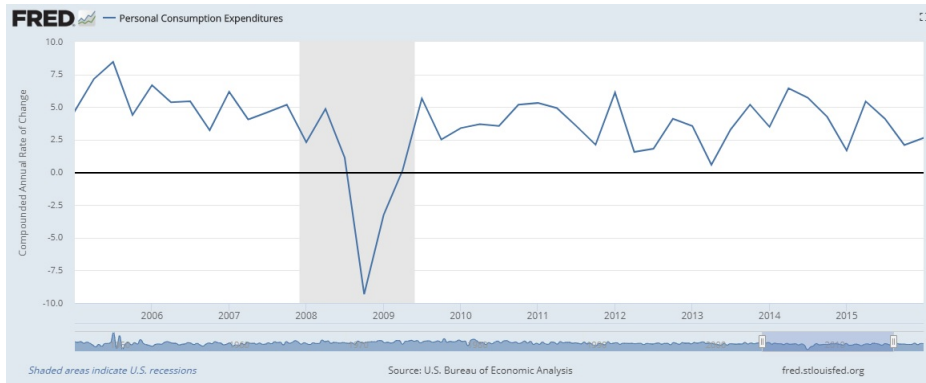


- Stock market trough in March 2009 at **50%** of Jan. 2007 peak value

Housing Price



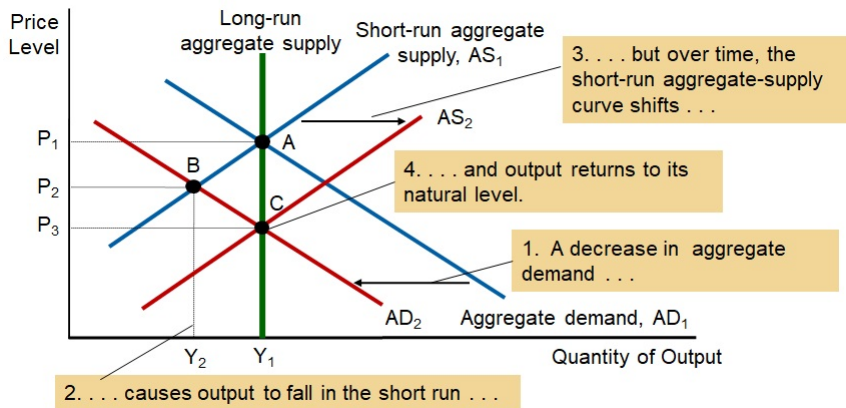
Consumption



Why Might a Fall in Asset Prices Reduce Consumer Spending

- *Reduction in wealth* makes consumers feel poorer.
- Lowers consumer confidence.
 - ▶ Fall in asset prices makes consumers pessimistic
- Stock price volatility causes uncertainty and leads to “wait and see” behavior.

Causes of Economic Fluctuations



Causes of Economic Fluctuations

The Self-Correcting Economy

- The economy tends to be self-correcting **in the long run**
 - ▶ Expansionary gap ($Y > Y^*$) $\Rightarrow$ price rises $\Rightarrow Y \downarrow$
 - ▶ Recessionary gap ($Y < Y^*$) $\Rightarrow$ price falls $\Rightarrow Y \uparrow$
- Why do we need policy to stabilize output?
 - ▶ It depends on the *speed* of the adjustment
 - ▶ If self-correction takes place *very slowly*, active use of policy can help to stabilize output

Causes of Economic Fluctuations

- Shift in aggregate supply
 - ▶ Firms – increase in production costs
 - ▶ Aggregate-supply curve: shifts left
- ▶ Short-run
 - ★ stagflation: Output falls, price level rises
- ▶ Long-run, if AD is held constant
- ▶ Short-run AS shifts back to right
- ▶ Output – natural level
- ▶ Price level - falls

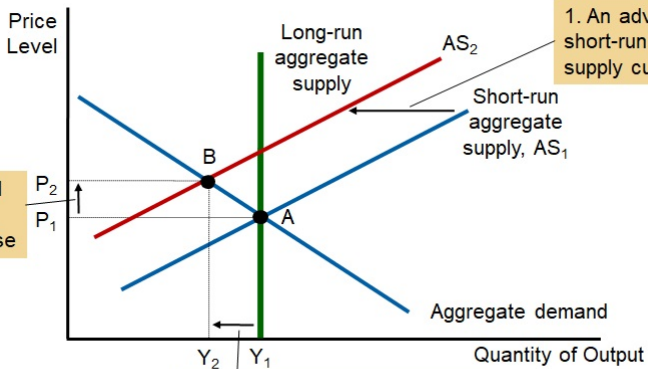
Example: Oil Shocks

- An inflation shock is a sudden change in the normal behavior of inflation, unrelated to the nation's output gap
 - ▶ An **adverse** inflation shock increases inflation
 - ▶ A **favorable** inflation shock decreases inflation
- In late 1973, the OPEC dramatically cut its supplies of crude oil
 - ▶ Oil prices was sharply increased
 - ▶ Prices of goods and services that were heavily dependent on oil increased
 - ▶ Increase in inflation rate

Example: Oil Shocks

- An inflation shock is a sudden change in the normal behavior of inflation, unrelated to the nation's output gap
 - ▶ An **adverse** inflation shock increases inflation
 - ▶ A **favorable** inflation shock decreases inflation
- In late 1973, the OPEC dramatically cut its supplies of crude oil
 - ▶ Oil prices was sharply increased
 - ▶ Prices of goods and services that were heavily dependent on oil increased
 - ▶ Increase in inflation rate

Causes of Economic Fluctuations



1. An adverse shift in the short-run aggregate-supply curve . . .

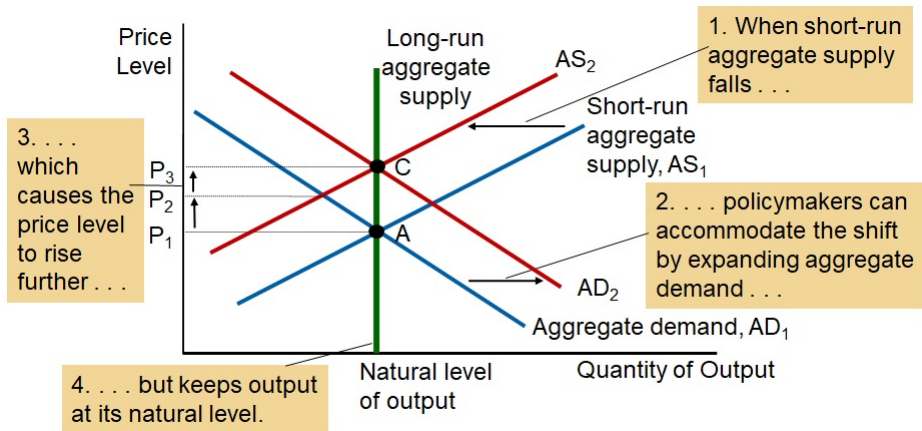
3. . . . and the price level to rise

2. . . . causes output to fall . . .

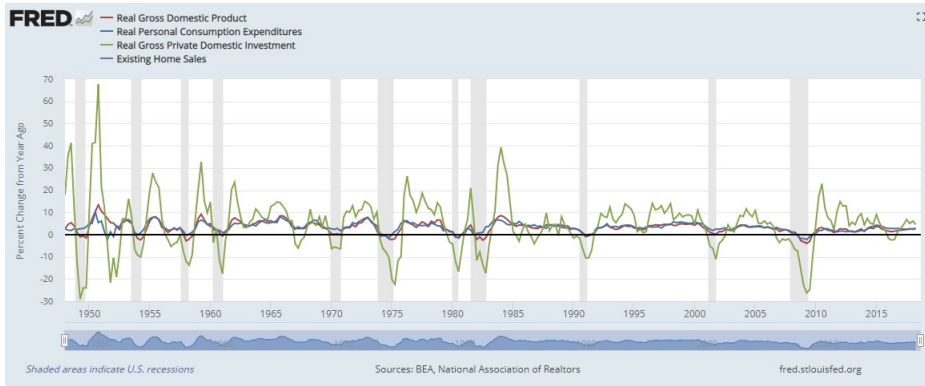
Causes of Economic Fluctuations

- Shift in aggregate supply
 - ▶ Firms – increase in production costs
 - ▶ Aggregate-supply curve: shifts left
- ▶ Short-run
 - ★ stagflation: Output falls, price level rises
- ▶ Long-run, *policymakers – shift AD to right*
- ▶ Output – natural level
- ▶ Price level – rises

Causes of Economic Fluctuations

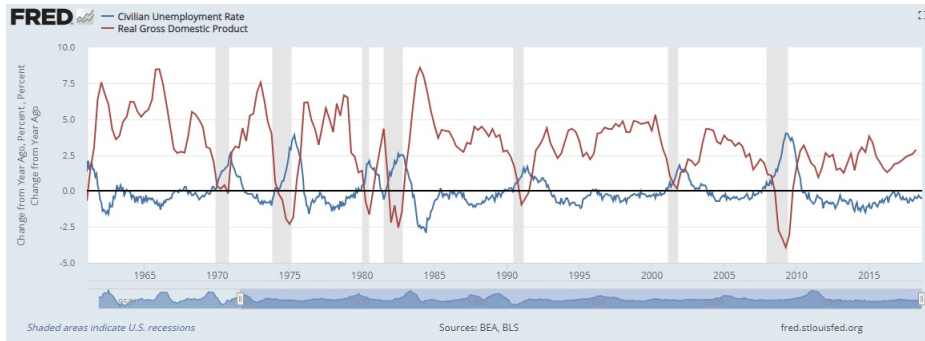


Procyclical Variables



Goback

Countercyclical Variable



Goback

Unemployment Rate



Source: U.S. Bureau of Labor Statistics via FRED®

Shaded areas indicate U.S. recessions.

fred.stlouisfed.org

Goback